Math 105L In Class 1 Practice

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Functions

1. Find the domain and range of the following functions:
   (a) $f(x) = \frac{1}{\sqrt{16-x^2}}$
   (b) $g(x) = \frac{3x-1}{x+2}$
   (c) $h(x) = \frac{4}{x^2+2}$
   (d) $k(x) = \sqrt{\frac{x-1}{x+3}}$ (Hint: You'll need to know where the fraction is positive/negative.)
   (e) $m(x) = 5 - |2x - 3|$
   (f) $p(x) = \sqrt{x^2 - 9}$
   (g) $r(x) = \sqrt{9 - |x|}$

2. Let $V = f(t)$ be the volume of gasoline (in gallons) remaining in a 15-gallon gas tank t hours after starting a road trip. The tank starts full, and the car runs out of gas after 5 hours.
   (a) What are the units of the domain and the range of f?
   (b) What is a suitable domain and range for f?
   (c) Describe in words what $f(2) - 3$ represents.

3. Let $T = g(m)$ be the temperature of an oven (in °F) m minutes after it is turned on to preheat.
   (a) Describe in words what $g(10)$ means, including units.
   (b) Describe in words the difference between $g(m + 15)$ and $g(m) + 15$.
   (c) Write an expression for a temperature that is 20 degrees Fahrenheit lower than twice the oven temperature 5 minutes ago.

4. A concert venue with a maximum capacity of 500 seats models its ticket sales using the function $N = C(p)$, where N is the number of tickets sold when the price per ticket is p dollars. Tickets are always priced between \$20 and \$100.
   (a) What would be a suitable domain for $C(p)$, including units?
   (b) What would be a suitable range for $C(p)$, including units and numerical type?
   (c) Describe in words what $C(50)$ represents.

5. Let $A = h(t)$ be the altitude (in meters) of a weather balloon t minutes after its launch.

   (a) Describe in words what $h(12)$ represents, including units.

   (b) Describe in words what $h(t - 10)$ represents.

   (c) Write an expression for an altitude that is 100 meters higher than half the balloon's altitude 15 minutes from now.

6. A coffee shop calculates its daily revenue R (in dollars) as a function of n, the number of cups of coffee sold on that day, given by $R = R(n)$. The shop can make at most 600 cups of coffee per day, and each cup costs \$4.

   (a) What is a sensible domain for $R(n)$? What are its units?

   (b) What is a sensible range for $R(n)$? What are its units?

   (c) Write an expression for the revenue earned from selling 30 more cups of coffee than on a day when n cups were sold.

7. You leave your office at 5:00 pm to drive home. For the first few minutes, you drive at a steady pace on local roads. You then get stuck in heavy stop-and-go traffic where you barely cover any distance. Once clear of traffic, you drive on the highway at a high, constant speed. Suddenly, you pull over at a gas station and stay there for a few minutes to fill up. Finally, you resume driving on the highway at a high speed until you reach home.
   Draw a graph of your distance from your office as a function of time. Take care to show distinct differences in slope between local driving, stop-and-go traffic, highway driving, and stopping for gas. Label your axes with variables and appropriate units.

8. You take your dog for a walk starting from your house. You walk at a relaxed, steady pace away from home. Your dog suddenly spots a squirrel and sprints straight ahead away from home for a short distance. You stop to catch your breath for two minutes. Realizing you went too far, you walk back halfway home at a steady pace, stop briefly to talk to a neighbor, and then jog the remaining distance back home.
   Draw a possible graph of your distance from home as a function of time. Label your axes with appropriate variables and units. Ensure differences between walking, sprinting, stopping, and jogging are clear.

9. You leave your home on a bicycle to go to the public library. You ride at a fast, constant speed until you arrive at the library. You spend 30 minutes inside studying. Upon leaving, you start heading back home at a slow, leisurely ride. Halfway home, you get a flat tire, so you stop for a few minutes to inspect it, and then walk the rest of the way home pushing your bike.
   Draw a possible graph of your distance from home as a function of time. Take care to show the differences in speed between riding fast, riding slow, stopping, and walking. Label your axes with variables and units.

10.

(a) For the relationship $p - q^3 = 1$, is p a function of q? If so, what is the range? If not, why not?

(b) For the relationship $p + |q| = 5$, is p a function of q? If so, what is the range? If not, why not?

(c) For the relationship $p^2 - q = 4$, is p a function of q? If so, what is the range? If not, why not?

(d) For the relationship $|p| + q = 3$, is p a function of q? If so, what is the range? If not, why not?

Linear Functions

1. (a) Suppose $h(x)$ is a linear function. Find its equation given that its slope is -4 and that $h(3) = 2$.

   (b) Suppose $k(x)$ is a linear function. Find its equation given that $k(-1) = 7$ and $k(3) = -1$.

   (c) Suppose that the line L is parallel to $y = -\frac{3}{4}x + 5$ and passes through the point $(-2, 4)$. Find its equation.

   (d) Find an equation for the line L that is perpendicular to $y = 2x - 3$ and passes through the point $(6, -1)$.

   (e) Find an equation for the line L that passes through the point $(4, 1)$ and is perpendicular to the line connecting $(1, 2)$ and $(3, 8)$.

2. A school district models the average annual paper usage P (in reams of paper) per classroom as a linear function of the number of digital assignment days d per month, given by

$$P = -4d + 80.$$

   If a district with 150 classrooms increases its digital assignment days by 3 days per month, how many total reams of paper does the model predict will be saved across the entire district over a 2-year period?

3. **The Strawberry Patch** In a commercial farm study, the following facts are known about growing strawberries:

   - If 15 thousand plants are planted per acre, each plant yields 2.4 pounds of strawberries per season.
   - For each additional thousand plants planted per acre, the yield decreases by 0.15 pounds per plant.

   Let x be the number of thousands of plants per acre, and let Y be the yield in pounds per plant.

   (a) What is Y if $x = 15$? Include units.

   (b) What information in the problem tells you that Y is a linear function of x? What is its slope?

   (c) Find a formula for Y in terms of x.

4. A municipal water authority models daily residential water usage W (in gallons per person) based on the municipal water rate r (in dollars per 1,000 gallons) using the linear function

$$W = -14.5r + 110.$$

   The county has a population of 45,000 residents. If the water authority raises the rate r by \$0.80 per 1,000 gallons, how many total gallons of water are expected to be conserved across the entire county during a 30-day month?

5. **Commuter Rail Ridership** A regional transit agency tracks ridership on a popular commuter line:

   - When the ticket price is \$12, the train averages 1,800 passengers per trip.
   - For every \$0.50 increase in ticket price, ridership drops by 40 passengers per trip.

   Let p be the ticket price in dollars, and let R be the number of passengers per trip.

   (a) What is R when $p = 12$? Include units.

   (b) Show that R is a linear function of p by finding its rate of change (slope). Include units for the slope.

   (c) Find a formula for R in terms of p.

6. A coffee roastery analyzes its carbon footprint and determines that roasting coffee at a temperature of 400°F results in 3.2 pounds of carbon emissions per bag of roasted coffee. For every 5°F reduction in roasting temperature, carbon emissions decrease by 0.15 pounds per bag.

   (a) Explain why the carbon emissions per bag, C (in pounds), is a linear function of roasting temperature T (in °F), and state its slope.

   (b) Find a formula for C in terms of T.

   (c) If the roastery reduces its roasting temperature from 400°F to 380°F, how many total pounds of carbon emissions will be prevented if they produce 25,000 bags of coffee this year?

7. (a) The electrostatic force F between two point charges is directly proportional to the product of their charges, q_1 and q_2, and inversely proportional to the square of the distance r between them. Write a formula for F in terms of q_1, q_2, and r.

   (b) The aerodynamic drag force D acting on a vehicle is directly proportional to the frontal surface area A of the vehicle and directly proportional to the square of its speed v. Write a formula for D in terms of A and v.

   (c) The electrical resistance R of a uniform wire is directly proportional to its length L and inversely proportional to its cross-sectional area A. Write a formula for R in terms of L and A.

   (d) The illuminance E on a surface produced by a light source is directly proportional to the luminous intensity I of the source and inversely proportional to the square of the distance d from the surface to the source. Write a formula for E in terms of I and d.

8. Consider the following *piecewise linear* function:

$$f(t) = \begin{cases} 2t+4, & t \leq 1, \\ -t+7, & t > 1. \end{cases}$$

   (a) Draw its graph.

   (b) Is $f(t)$ continuous at $t = 1$? Explain.

9. Consider the following *piecewise linear* function:

$$g(t) = \begin{cases} -t+2, & t < 2, \\ 2t-1, & t \geq 2. \end{cases}$$

   (a) Draw its graph.

   (b) Fill in the blank: At $t = 2$, the function $g(t)$ is not _____.

10. Consider the following *piecewise linear* function:

$$h(t) = \begin{cases} 3t+8, & t \leq -2, \\ -0.5t+1, & -2 < t < 2, \\ 2t-4, & t \geq 2. \end{cases}$$

   (a) Draw its graph.

   (b) Is $h(t)$ continuous everywhere? Explain what happens at $t = -2$ and $t = 2$.

Power functions and Polynomials

1. (a) **True or False:** A polynomial of degree 6 must have exactly 5 turning points. Explain your reasoning.

   (b) **True or False:** For the polynomial $p(x) = -3x^5 + 7x^2 - 1$, both tails of the graph point downwards toward $-\infty$ as $x \to \pm\infty$. Explain your reasoning.

   (c) **True or False:** If a polynomial has an even degree and a positive top coefficient, then $\lim_{x\to-\infty} p(x) = \infty$ and $\lim_{x\to\infty} p(x) = \infty$. Explain your reasoning.

   (d) **True or False:** Every cubic polynomial ($n = 3$) has at least one turning point. Explain your reasoning.

   (e) **True or False:** The graph of $q(x) = -2x^4 + 5x^3 - x + 4$ has tails that point in opposite directions because the top coefficient is negative. Explain your reasoning.

   (f) **True or False:** If a polynomial graph exhibits 4 turning points, the degree of the polynomial must be at least 5. Explain your reasoning.

2. Without using a calculator or online tools, sketch the graphs of the following polynomials. Be sure to clearly label all x- and y-intercepts.

   (a) $y = (x+2)(x-1)(x-3)$

   (b) $y = -(x+1)^2(x-2)$

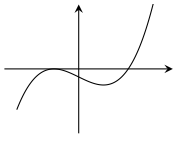
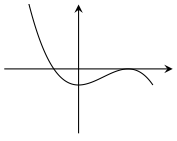
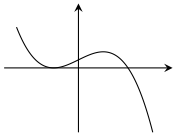
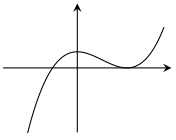
   (c) $y = \frac{1}{4}x(x+2)^2(x-2)$

   (d) $y = -x(x-2)^3$

   (e) $y = \frac{1}{3}(x+2)(x-1)^2(x-3)$

3. Without using a calculator or online tools, match each of the following polynomials with the correct graph.

$(x-2)^2(x+1)$	$-(x-2)^2(x+1)$	$(x-2)(x+1)^2$	$-(x-2)(x+1)^2$
(A)	**(B)**	**(C)**	**(D)**

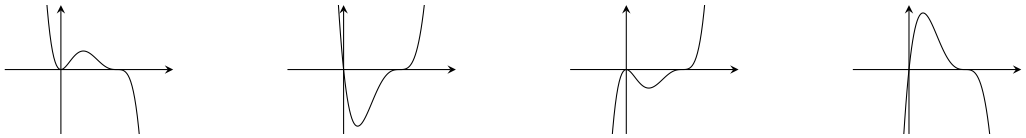
4. Without using a calculator or online tools, match each of the following polynomials with the correct graph.

$x^2(x-2)(x+2)$	$-x^2(x-2)(x+2)$	$x(x-2)^2(x+2)$	$-x(x-2)^2(x+2)$
(**A**)	(**B**)	(**C**)	(**D**)



5. Without using a calculator or online tools, match each of the following polynomials with the correct graph.

$x(x-1)^3$	$-x(x-1)^3$	$x^2(x-1)^3$	$-x^2(x-1)^3$
(**A**)	(**B**)	(**C**)	(**D**)



6. Without using a calculator or online tools: by first factoring, sketch the graphs of the following polynomials. Be sure to label all of the intercepts.

   (a) $p(x) = 4 - x^2$

   (b) $p(x) = x^3 - 4x$

   (c) $p(x) = x^3 - 2x^2 + x$

   (d) $p(x) = -\frac{1}{3}(x^4 - 9x^2)$

Rational Functions

1. Find all of the intercepts and asymptotes of the following functions.

   (a) $r(x) = \frac{2x}{x^2-9}$

   (b) $r(x) = \frac{3x-6}{x+2}$

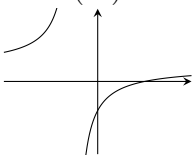
   (c) $r(x) = \frac{(x-2)(x+3)}{x-1}$

   (d) $r(x) = \frac{4-x^2}{x^2+1}$

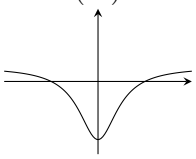
2. Match each graph below with its corresponding function. Note that one function will not have a matching graph.

$$f(x) = \frac{x^2-4}{x^2+1} \qquad g(x) = \frac{x-2}{x+1} \qquad h(x) = \frac{(x+2)(x-1)}{x-2} \qquad k(x) = \frac{1}{x^2-4}$$

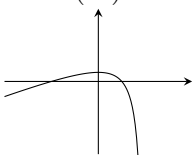
(A)



(B)



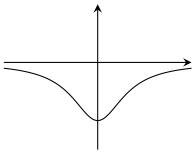
(C)



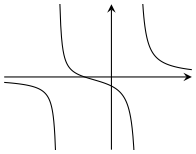
3. Match each graph below with its corresponding function. Note that one function will not have a matching graph.

$$f(x) = \frac{x+1}{(x-1)(x+2)} \qquad g(x) = \frac{(x+1)(x-1)}{x} \qquad h(x) = \frac{-2}{x^2+1} \qquad k(x) = \frac{x^2+1}{x^2-1}$$

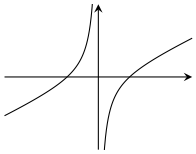
(A)



(B)



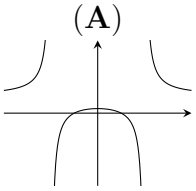
(C)



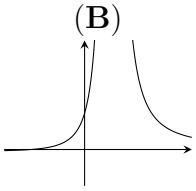
4. Match each graph below with its corresponding function. Note that one function will not have a matching graph.

$$f(x) = \frac{x-3}{x-1} \qquad g(x) = \frac{x^2-1}{x^2-4} \qquad h(x) = \frac{x+2}{(x-1)^2} \qquad k(x) = \frac{2x+1}{x+2}$$

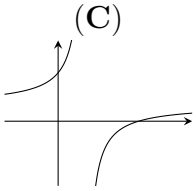
(A)



(B)



(C)



5. Consider the rational function

$$r(x) = \frac{(x-3)(x+1)}{(x-2)(x+2)}.$$

Perform a complete analysis of $r(x)$ by finding its intercepts, asymptotes, and sign behavior, then sketch its graph. Do not use a calculator or any graphing software to complete this problem.

6. Consider the rational function

$$r(x) = \frac{(2x+3)(x-1)}{(x+1)(x-3)}.$$

Perform a complete analysis of $r(x)$ by finding its intercepts, asymptotes, and sign behavior, then sketch its graph. Do not use a calculator or any graphing software to complete this problem.

7. Consider the rational function

$$r(x) = \frac{(x-1)(x+2)(x-3)}{(x+1)(x-2)}.$$

Perform a complete analysis of $r(x)$ by finding its intercepts, asymptotes, and sign behavior, then sketch its graph. Do not use a calculator or any graphing software to complete this problem.

8. Consider the rational function

$$r(x) = \frac{(x+2)(x-2)}{x(x+1)^2}.$$

Perform a complete analysis of $r(x)$ by finding its intercepts, asymptotes, and sign behavior, then sketch its graph. Do not use a calculator or any graphing software to complete this problem.

9. Classify each of the following functions by listing all applicable labels from: **Power function**, **Polynomial**, and **Rational function**.

   (a) $f(x) = 7x^5$

   (b) $f(x) = 4x^{-3}$

   (c) $f(x) = 3x^{2.5}$

   (d) $f(x) = 2x^4 - 5x^2 + 1$

   (e) $f(x) = \frac{x^2-4}{x+3}$

   (f) $f(x) = \frac{5}{x^3+2x}$

   (g) $f(x) = \frac{1}{2}x^8$

   (h) $f(x) = -9$

   (i) $f(x) = 6x^{-1/2}$

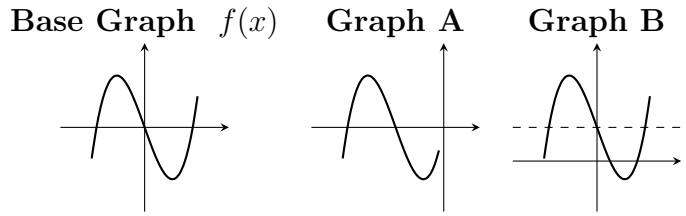
   (j) $f(x) = \frac{3x^3+1}{4}$

Function Transformations

1. The graph of a function $f(x)$ is shown below. Consider the following three transformation formulas:

$$\text{(i) } y = f(x+2), \qquad \text{(ii) } y = f(x-2), \qquad \text{(iii) } y = f(x) + 2$$

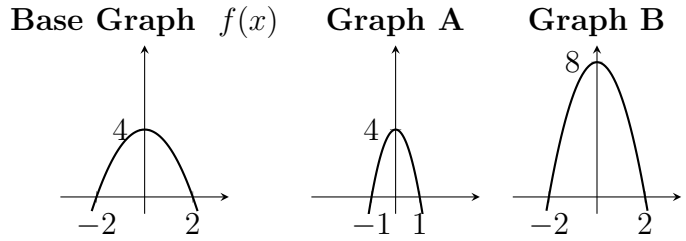
Match Graph A and Graph B with their corresponding formulas. (One formula will not match either graph.)



2. The graph of a function $f(x)$ is shown below. Consider the following three transformation formulas:

$$\text{(i) } y = f(2x), \qquad \text{(ii) } y = 2f(x), \qquad \text{(iii) } y = f\left(\frac{1}{2}x\right)$$

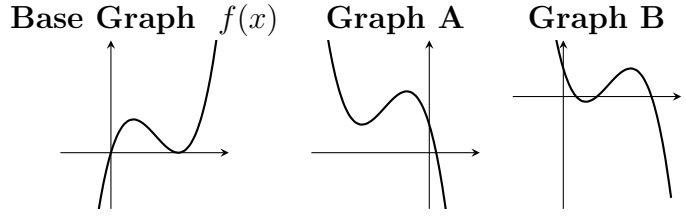
Match Graph A and Graph B with their corresponding formulas. (One formula will not match either graph.)



3. The graph of a function $f(x)$ is shown below. Consider the following three transformation formulas:

$$\text{(i) } y = -f(x) + 1, \qquad \text{(ii) } y = f(-x) + 1, \qquad \text{(iii) } y = -f(x+1)$$

Match Graph A and Graph B with their corresponding formulas. (One formula will not match either graph.)

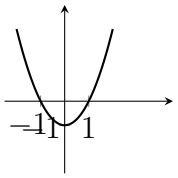


4. The graph of a function $f(x)$ is shown below. Consider the following three transformation formulas:

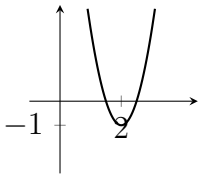
$$\text{(i) } y = f(2x-4), \qquad \text{(ii) } y = f(2x+4), \qquad \text{(iii) } y = 2f(x-2)$$

Match Graph A and Graph B with their corresponding formulas. (One formula will not match either graph.)

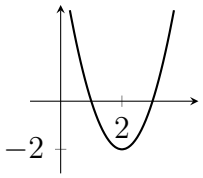
Base Graph $f(x)$



Graph A

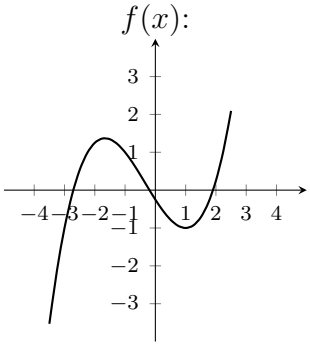


Graph B

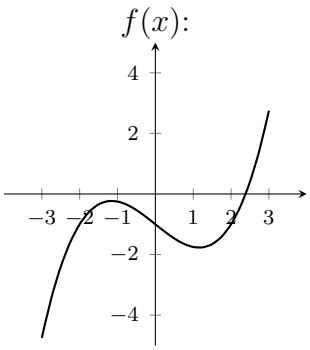


5. For each problem below, graph the given transformation of $f(x)$. If multiple transformations are involved, draw each step on a separate set of axes, as well as your final answer.

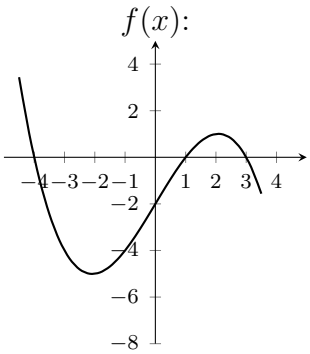
(a) Given the graph of $f(x)$ below, graph the transformed function $g(x) = f(x-2)+1$.



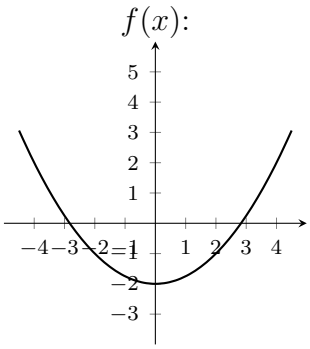
(b) Given the graph of $f(x)$ below, graph the transformed function $g(x) = -2f(x)$.



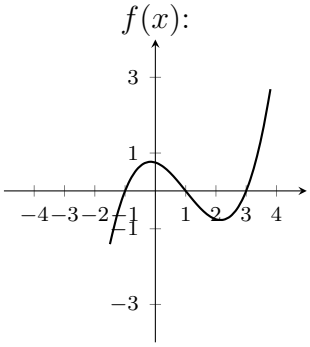
(c) Given the graph of $f(x)$ below, graph the transformed function $g(x) = f(-x) - 2$.



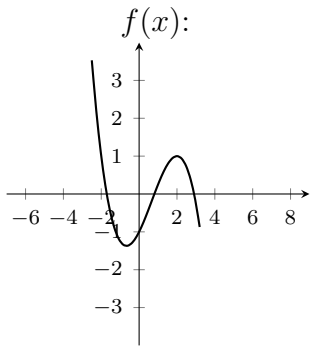
(d) Given the graph of $f(x)$ below, graph the transformed function $g(x) = f(2x) + 3$.



(e) Given the graph of $f(x)$ below, graph the transformed function $g(x) = -f(x+3)$.

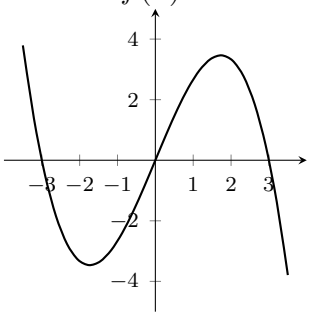


(f) Given the graph of $f(x)$ below, graph the transformed function $g(x) = f\left(\frac{1}{2}x - 1\right)$.



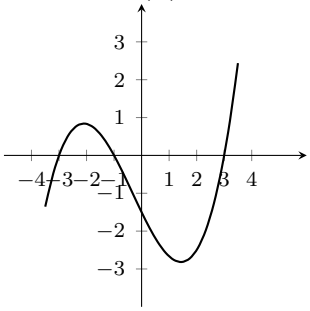
(g) Given the graph of $f(x)$ below, graph the transformed function $g(x) = -\frac{1}{2}f(x)+2$.

$f(x)$:



(h) Given the graph of $f(x)$ below, graph the transformed function $g(x) = f(-x+2)$.

$f(x)$:



Modeling

1. Suppose a person enters medical school and specialized residency training at age 20, completes all training at age 32, and practices medicine until retiring at age 62. Let x represent the person's age in years ($0 \leq x \leq 62$).

   Write a piecewise formula for $F(x)$, the proportion of their total lifetime lived so far that they spent in medical training. Then sketch a graph of $F(x)$ vs. x.

2. A suburban township with 10,000 residential homes launches a community solar adoption initiative. Market research reveals that:

   (i) 10% of the homes have heavily shaded roofs and can never install solar panels,

   (ii) at time $t = 0$ months, exactly 50 early-adopter homes have solar panels installed,

   (iii) the promotion campaign runs for 36 months, after which adoption levels off.

   Draw a possible graph of $N(t)$, the total number of homes with solar panels t months after the launch. Label your axes with appropriate scales and write a short paragraph justifying the shape of your graph.

3. A community climbing gym offers a pay-per-visit rate of \$15 per visit. Alternatively, climbers can purchase a monthly pass for a flat fee of \$40 plus a reduced rate of \$5 per visit.

   (a) Calculate the total monthly cost under both options if a climber visits 2 times in a month. What if they visit 8 times?

   (b) Write a formula for the monthly savings $S(v)$ (in dollars) gained by purchasing the monthly pass instead of paying per visit, as a function of the number of visits v.

   (c) Find the minimum number of visits v in a month required for the monthly pass to be more cost-effective than paying per visit.

4. To estimate the height of a tall oak tree in a park, a student measures the length of the tree's shadow on level ground to be 28 feet. At the exact same time, a 5-foot tall wooden fence post nearby casts a shadow of length 4 feet. Draw a sketch representing this situation, and find the height of the oak tree.

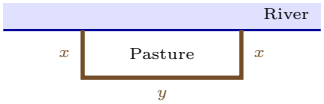
5. The illuminance I (measured in lux) produced by a light bulb on a surface is inversely proportional to the square of the distance d (in meters) from the light source.

   (a) Write a general formula expressing I in terms of d.

   (b) If the illuminance is 180 lux at a distance of 2 meters, find the constant of proportionality k and state its units.

   (c) Determine the illuminance when the distance from the light bulb is increased to 6 meters.

6. A farmer wants to construct a rectangular pasture along a straight riverbank (as shown below). No fencing is required along the river. The farmer has 160 meters of fencing material available for the remaining three sides.



   (a) Let x be the length (in meters) of the two sides perpendicular to the river. Express the length y of the side parallel to the river in terms of x.

   (b) Write a formula for $A(x)$, the enclosed area of the pasture, as a function of x.

   (c) Find the dimensions x and y that maximize the area of the pasture, and calculate the maximum area. (No calculus here: just use symmetry.)

7. A comedy club with a capacity of 300 seats charges \$25 per ticket and consistently sells out every show. Market research indicates that for every \$1 increase in ticket price above \$25, attendance decreases by 10 people.

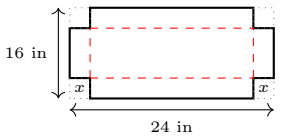
   (a) Find a linear formula for $N(p)$, the number of attendees, as a function of the ticket price p (where $p \geq 25$).

   (b) Write a formula for total revenue $R(p)$ per show as a function of ticket price p.

   (c) Determine the ticket price p that maximizes revenue per show, find the maximum total revenue, and sketch a graph of $R(p)$ vs. p. (No calculus here: just use symmetry.)

8. **Challenge Problem**

   An open-top box is to be constructed from a rectangular piece of cardboard measuring 16 inches by 24 inches by cutting equal squares of side length x inches from each of the four corners and folding up the resulting flaps (as shown below).



   (a) Write expressions for the height, width, and length of the box in terms of x.

   (b) Write a formula for $V(x)$, the volume of the box (in cubic inches), as a function of x.

   (c) State the practical physical domain for x in this context.

   (d) Draw the entire graph of $V(x)$, highlight the portion corresponding to the practical domain, and circle the point where $V(x)$ is maximized.

Rates of Change and Derivative at a Point

For all problems in this section, you may not use any calculus tools beyond what was covered up until Worksheet 4-3. For example, using the power rule, product rule, quotient rule, or chain rule is not allowed.

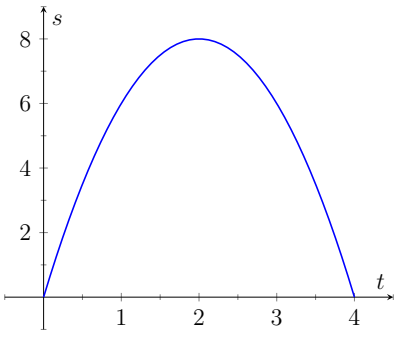
1. The position $s(t)$ (in feet) of a vehicle traveling along a straight track is recorded at various times t (in seconds) in the table below:

t (sec)	0	1.0	2.0	3.0	4.0
$s(t)$ (ft)	10.0	24.5	51.0	89.5	140.0

   (a) Calculate the average velocity of the vehicle between $t = 1.0$ and $t = 3.0$ seconds.

   (b) Calculate the average velocity of the vehicle between $t = 2.0$ and $t = 4.0$ seconds.

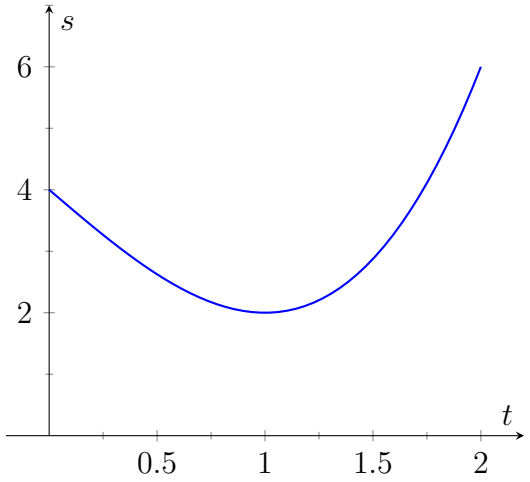
2. Given the graph of the position function $s(t) = -2(t-2)^2 + 8$ below, draw and label lines whose slopes correspond to the average velocity over the specified time intervals:



   (a) Average velocity on the interval $[0, 2]$.

   (b) Average velocity on the interval $[1, 3]$.

3. Given the graph of the position function $s(t) = t^3 - 3t + 4$ below, draw and label lines whose slopes correspond to the average velocity over the specified time intervals:



   (a) Average velocity on the interval $[0, 1]$.

   (b) Average velocity on the interval $[0, 2]$.

4. The table below provides values of the average velocity $\frac{s(3+h)-s(3)}{h}$ of an object near $t = 3$ seconds:

h (sec)	Avg. Vel on $[3, 3+h]$ (m/s)	h (sec)	Avg. Vel on $[3, 3+h]$ (m/s)
-0.1	5.90	0.1	6.10
-0.01	5.99	0.01	6.01
-0.001	5.999	0.001	6.001

   Estimate the instantaneous velocity of the object at $t = 3$ seconds.

5. The table below provides values of the average velocity $\frac{s(1+h)-s(1)}{h}$ of an object near $t = 1$ second:

h (sec)	Avg. Vel on $[1, 1+h]$ (ft/s)	h (sec)	Avg. Vel on $[1, 1+h]$ (ft/s)
-0.1	-2.85	0.1	-3.15
-0.01	-2.985	0.01	-3.015
-0.001	-2.9985	0.001	-3.0015

   Estimate the instantaneous velocity of the object at $t = 1$ second.

6. For each of the following position functions $s(t)$ (where s is in meters and t is in seconds), compute the exact instantaneous velocity at the given time using the limit definition $v(a) = \lim_{h \to 0} \dfrac{s(a+h) - s(a)}{h}$.

   (a) $s(t) = 2t^2 - 3t$ at $t = 2$.

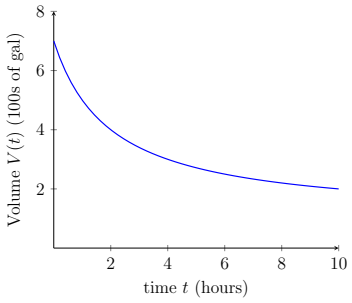
   (b) $s(t) = -t^2 + 5t + 1$ at $t = 3$.

   (c) $s(t) = 3t^2 + 4$ at $t = 1$.

(d) $s(t) = t^3$ at $t = 2$.

(e) $s(t) = \frac{2}{t}$ at $t = 2$.

7. A water storage tank is being drained for maintenance. The volume of water remaining in the tank (in hundreds of gallons) t hours after draining begins is modeled by $V(t) = \frac{12}{t+2} + 1$.



(a) How many hundreds of gallons of water were drained during the first 2 hours? What was the average rate of draining per hour over this period?

(b) Illustrate the average rate of change on the graph above.

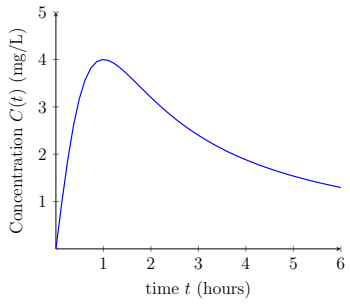
(c) Rank the following quantities in order from smallest to largest:

(i) Instantaneous rate of change of volume at $t = 0$.

(ii) Average rate of change of volume between $t = 0$ and $t = 2$.

(iii) Instantaneous rate of change of volume at $t = 2$.

8. The concentration of a medication in a patient's bloodstream (in mg/L) t hours after ingestion is modeled by $C(t) = \frac{8t}{t^2+1}$.



(a) Find the average rate of change of medication concentration over the first 3 hours ($[0, 3]$).

(b) Illustrate the average rate of change on the graph above.

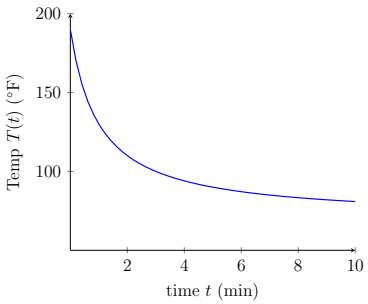
(c) Rank the following quantities in order from smallest to largest:

(i) Instantaneous rate of change at $t = 1$.

(ii) Average rate of change on $[0, 3]$.

(iii) Instantaneous rate of change at $t = 3$.

9. The temperature (in °F) of a freshly baked pie placed on a cooling rack t minutes after coming out of the oven is given by $T(t) = 70 + \frac{120}{t+1}$.



(a) Find the average cooling rate (in °F/min) of the pie over the first 3 minutes.

(b) Illustrate the average cooling rate on the graph above.

(c) Estimate $T'(3)$ using $h = 0.001$.

10. Consider the function $f(x) = x^2 - 4x$.

    (a) Fill in the table below to estimate $f'(3)$:

h	Avg. RoC on $[3, 3+h]$	h	Avg. RoC on $[3, 3+h]$
-0.1		0.1	
-0.01		0.01	
-0.001		0.001	

    (b) Find $f'(3)$ exactly using the limit definition of the derivative.

    (c) Find the equation of the line tangent to the graph of $y = f(x)$ at $x = 3$.

11. Consider the function $f(x) = -2x^2 + 5$.

    (a) Fill in the table below to estimate $f'(1)$:

h	Avg. RoC on $[1, 1+h]$	h	Avg. RoC on $[1, 1+h]$
-0.1		0.1	
-0.01		0.01	
-0.001		0.001	

    (b) Find $f'(1)$ exactly using the limit definition of the derivative.

    (c) Find the equation of the line tangent to the graph of $y = f(x)$ at $x = 1$.

12. Consider the function $f(x) = x^3 - 2x$.

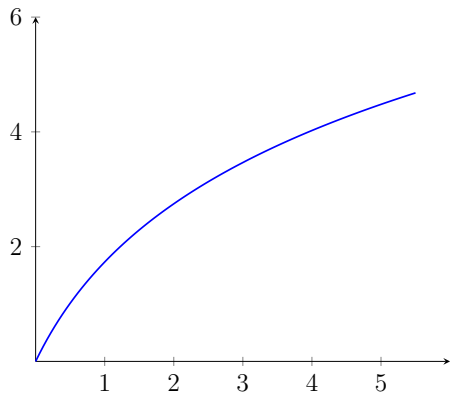
    (a) Fill in the table below to estimate $f'(2)$:

h	Avg. RoC on $[2, 2+h]$	h	Avg. RoC on $[2, 2+h]$
-0.1		0.1	
-0.01		0.01	
-0.001		0.001	

    (b) Find $f'(2)$ exactly using the limit definition of the derivative.

    (c) Find the equation of the line tangent to the graph of $y = f(x)$ at $x = 2$.

13. For the function $f(x)$ graphed below, compare the following pairs of quantities. Indicate which is larger ($>$ or $<$), and illustrate each pair on a copy of the graph:

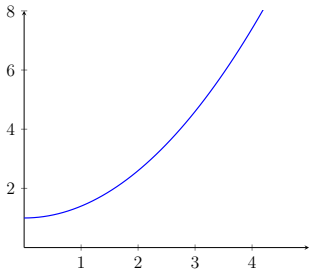


(a) The average rate of change on $[1, 3]$ versus the average rate of change on $[3, 5]$.

(b) $f(2)$ versus $f(4)$.

(c) $f'(1)$ versus $f'(4)$.

14. For the function $g(x)$ graphed below, compare the following pairs of quantities. Indicate which is larger ($>$ or $<$), and illustrate each pair on a copy of the graph:

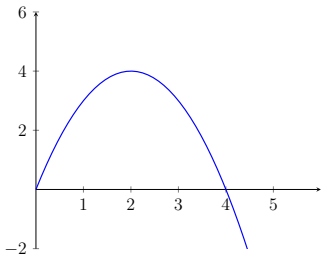


(a) The average rate of change on $[1, 2]$ versus the average rate of change on $[2, 4]$.

(b) $g'(1)$ versus $g'(3)$.

(c) $g(1)$ versus $g(3)$.

15. For the function $h(x)$ graphed below, compare the following pairs of quantities. Indicate which is larger ($>$ or $<$), and illustrate each pair on a copy of the graph:



(a) The average rate of change on $[1, 2]$ versus the average rate of change on $[2, 4]$.

(b) $h'(1)$ versus $h'(3)$.

(c) $h'(2)$ versus $h'(4)$.